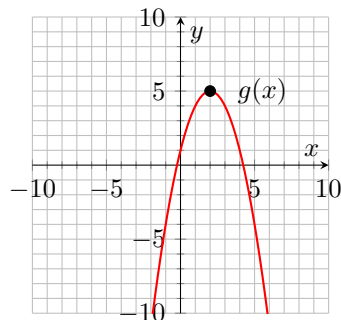


You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

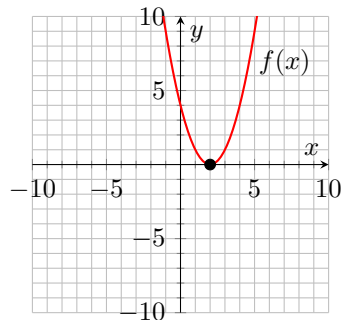
1. [2pts each] Use the graph of the quadratic function, to answer the following:

- What is the vertex?
 $(2, 5)$
- What is the axis of symmetry?
 $x = 2$
- What is the range?
 $(-\infty, 5]$
- What is the domain?
 $(-\infty, \infty)$
- Using the graph, what interval(s) is the function positive? (approximate)
 $\approx (-.24, 4.24)$



2. Using the function $f(x) = x^2 - 4x + 4$, complete the following:

- [2pts] Determine f 's axis of symmetry.
 $f(x) = (x - 2)^2$ so the vertex is $(2, 0)$ and thus the axis of symmetry is, $x = 2$. (Alternatively, use $x = -\frac{b}{2a} = -\frac{-4}{2} = 2$).
- [2pts] If any exist, find the x -int(s).
 $x = 2$, where the vertex touches the axis.
- [2pts] Find the y -int.
 $f(0) = 4$, so $y = 4$.
- [4pts] Sketch the graph



3. Consider $P(x) = x^2 - 5x + 2$ and $(x - 2)$.

- [4pts] Use division (synthetic or traditional) to find the quotient and remainder when $P(x)$ is divided by $x - 2$.

$$\begin{array}{r|rrr} 2 & 1 & -5 & 2 \\ & & 2 & -6 \\ \hline & 1 & -3 & -4 \end{array}$$

Quotient, $Q(x) = x - 3$ and the remainder, $r = -4$.

- [4pts] Find $\frac{P(x)}{x - 2}$

$$\begin{aligned} \frac{P(x)}{x - 2} &= Q(x) + \frac{r}{x - 2} \\ &= x - 3 + \frac{-4}{x - 2} \end{aligned}$$

- [2pts] Is $x = 2$ a root/zero of $P(x)$?

No. Since the remainder is -4 , $P(2) = -4$. 2 is a root/zero only if $P(2) = 0$.

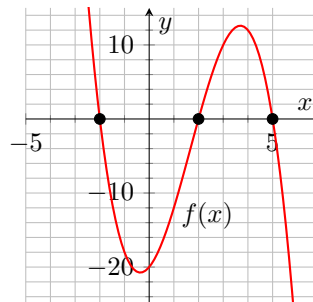
4. Consider the polynomial $f(x)$ shown in the graph.

a. What are the zeros of $f(x)$?

$x = -2, 2, 5$; when it crosses the x -axis.

b. Find the simplest polynomial with real coefficients that has the roots from above (you do not have to expand).

$$P(x) = (x + 2)(x - 2)(x - 5)$$



5. Use $f(x) = x^3 - 4x$ to complete the following.

a. [3pts] Find the x -int(s)

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

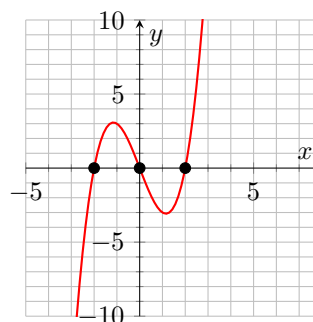
$$x(x - 2)(x + 2) = 0$$

So $x = 0, 2, -2$.

b. [3pts] Determine if the graph crosses or touches without crossing the x -axis at each x -int.

The multiplicity of each zero is odd; so it crosses at each zero.

c. [4pts] Sketch a graph of the function.



6. [5pts each] Find all real solutions.

a. $\sqrt{x+3} = x+1$

$$(\sqrt{x+3})^2 = (x+1)^2$$

$$x+3 = x^2+2x+1$$

$$0 = x^2+x-2$$

$$0 = (x+2)(x-1)$$

So $x = -2, 1$. (check that these are solutions)

b. $x^{-2/5} = 4$

$$(x^{-2/5})^{-5/2} = (4)^{-5/2}$$

$$x = ((4)^{1/2})^{-5}$$

$$x = \frac{1}{2^5} = \frac{1}{32}$$

7. [5pts each] Find the vertex of the graph of the following functions:

a. $f(x) = x^2 - 4x + 10$.

$$a = 1 \text{ and } b = -4. \text{ Use vertex} = \left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right) \right) = (2, f(2)) = (2, 6).$$

b. $f(x) = -3(x+2)^2 - 5$.

Use vertex = $(h, k) = (-2, -5)$

8. Use $f(x) = \frac{2x+6}{x-2}$

a. [4pts] Find the domain of $f(x)$.

$x \neq 2$, so domain: $(-\infty, 2) \cup (2, \infty)$

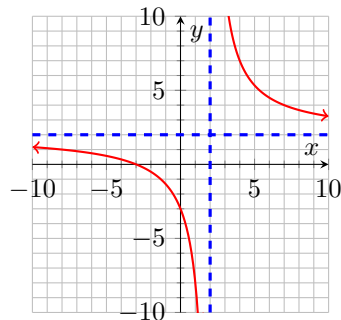
b. [4pts] Find all vertical asymptotes.

$x = 2$

c. [6pts] Find the horizontal asymptote.

$$\lim_{x \rightarrow \infty} \frac{2x+6}{x-2} \cdot \frac{1/x}{1/x} = \frac{2+6/x}{1-2/x} = 2$$

d. [6pts] Sketch a graph of the function



9. [2pts each] Find the following:

a. $x^2 - 20$ as $x \rightarrow -\infty$

∞ since x^2 is even and $1 > 0$.

b. $-2x^{13} + x^4 - 50$ as $x \rightarrow \infty$

$-\infty$ since x^{13} is odd and $-2 < 0$.

c. $\lim_{x \rightarrow \infty} \frac{1}{x^3}$
0

d. $\lim_{x \rightarrow 0^+} \frac{3}{x}$

∞ ($0^+ \Rightarrow$ it approaches 0 from the right; so it stays positive)

e. $\lim_{x \rightarrow 2^-} \frac{3}{x-2}$

$-\infty$ ($2^- \Rightarrow$ it approaches 2 from the left; so it stays negative)

f. The domain of $\frac{x^2 - 3x - 7}{x(x-2)(x+1)}$

$x \neq 0, 2, -1$. So domain: $(-\infty, -1) \cup (-1, 0) \cup (0, 2) \cup (2, \infty)$.